Digestion and Renal Sciences

SGT3: Liver

Consider the following case history of a patient with gallstones. You should revise your lectures on the subject and read around the relevant topics in your textbooks. The points listed after the case history should guide your studying. You will be expected to discuss this case at the appropriate SGT / PBL B.

JAUNDICE

An obese middle-aged woman visited her GP and complained of feeling unwell. She said she suffered bouts of severe pain in the upper abdomen. The pain became more severe during and after meals. The woman had a yellowish (jaundiced) appearance. She also said her faeces were pale, straw coloured and floated in the toil

et pan. The GP examined the woman. He suspected that she was suffering from gallstones and she was referred for further investigations.

1. Symptoms :
   Why did the severity of the pain increase during a meal?

2. Gallstones :
   (i) What is the composition of the two main types of gallstone?
   (ii) What could be the cause of development of each type of gallstone?

3. Bilirubin :
   (i) Jaundice is due to the accumulation of bilirubin in the blood. In this case would the bilirubin be mainly conjugated or unconjugated?
   (ii) Which condition(s) are characterised by accumulation of <u>conjugated</u> bilirubin in the blood?
   (iii) Which are characterised by accumulation of <u>unconjugated</u> bilirubin in the blood?
   (iv) How is bile pigment transported into the hepatocytes?
   (v) How are they metabolised in the liver?
   (vi) How are they transported into the bile?

4. Diagnosis and Treatment
    (i) How is gallstone disease likely to be diagnosed in hospital?
    (ii) How is gallstone disease likely to be treated non-invasively?

5. Faecal Abnormality
   Explain the abnormal appearance of the woman's faeces and why they float?

6. Cholecystectomy
   What would be the physiological consequences of treating this condition by removal of the gall bladder (cholecystectomy)?